

Factorization Exercises

1 1 Basic Level (Common factor, difference of squares)

Factorize each of the following polynomials:

1. $3x + 6$
2. $5a^2 - 10a$
3. $x^2 - 9$
4. $4y^2 - 25$
5. $6m^2 + 9m$

2 2 Intermediate Level (Perfect square trinomial, quadratic trinomial)

1. $x^2 + 6x + 9$
2. $a^2 - 8a + 16$
3. $x^2 + 5x + 6$
4. $x^2 - 2x - 8$
5. $2x^2 + 7x + 3$

3 3 Advanced Level (Grouping method, repeated factorization)

1. $ax + ay + bx + by$
2. $x^3 - 4x$
3. $x^2 - y^2 + x + y$
4. $3x^2 - 12xy + 12y^2$
5. $x^4 - 16$
6. $x^3 + 3x^2 - 4x - 12$